INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION
FOR THE
TESTER, CONSTRUCTION MATERIALS, MOISTURE & DENSITY
(TCMMD)

1. SCOPE. This commercial item description covers a single unit density and moisture tester with radioactive sources. The tester is used for measuring the density and moisture content of soils, and the density of asphalt pavement and other construction material.

2. SALIENT CHARACTERISTICS. The TCMMD will be a one-piece self contained instrument capable of measuring the density of soils, asphalt pavement, and other construction material. The TCMMD will be capable of non-destructively measuring the moisture content of soils. It's desired that the TCMMD be capable of measuring the density of soils, asphalt pavement, and other construction materials and measuring the moisture content of soils through non-destructive methods. The TCMMD shall be a measuring device comprised of radioactive sources, radioactive detectors, a readout device, data storage, and an integral power supply. The TCMMD shall be capable of testing for density and moisture by backscatter, direct transmission, and (optional) air gap methods.

2.1 Radioactive sources. The sources shall be doubly or triply encapsulated and sealed having sealed source certification. The radiological surface dose rate shall not be greater than 40.00 millirem/hour (mrem/hr) at the surface of the TCMMD (neutron and gamma sources).

2.2 Radioactive detectors. The detectors shall be any suitable type of radioactive detectors.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: US Army Geospatial Center, 7701 Telegraph Road, ATTN: CEAGC-GST, Alexandria, VA 22315-3864 or emailed to thomas.coxjr@us.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

AMSC N/A FSC 6635

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2.3 Measurement Depth. The TCMMD shall be capable of depth measurements up to 12 inches in 2 inch increments or less. The index rod or index strip on the gauge allowing the trigger mechanism to be latched or locked at the desired measurement depth. The gauge being designed so that if the operator attempts to pick it up without first returning the source rod to the shielded position, the trigger in the handle releases and the rod retracts to the shielded position. The source rod must be lockable.

2.4 Visual display. The TCMMD shall have a self contained, easily readable and understandable visual display. The visual display shall provide an English readout with automatic conversion to metric units. The visual display shall not become obscured while operating in humid, wet, cold, hot, dusty, or dry environments. The visual display shall incorporate an integrated back light that enables the analyst to read the visual display in reduced visibility, low light, conditions. The visual display shall provide a direct readout of Wet Density (WD) in lbs/ft^3, Dry Density (DD) in lbs/ft^3, moisture content in lbs/ft^3, percent moisture, percent air void, percent proctor, percent marshall, and percent compaction. The TCMMD visual display shall also display the remaining power supply capacity. At a minimum, the visual display shall contain counting circuits, timing circuits, a data display, and related electronic components.

2.5 Data storage and Data transfer. The TCMMD Shall have the ability to internally record a minimum of 120 test records and be provided with a Universal Serial Bus (USB) port to allow data to be transferred from the TCMMD onto external data storage devices. Each record shall include the date conducted and provision for entering a task identifier of not less than five characters.

2.6 Power supply. The TCMMD shall be provided with an operator removable, rechargeable, replaceable, extended life battery source capable of performing a minimum of 3000 one minute tests before recharging is needed. The TCMMD shall be provide with a backup power supply that allows the operator to perform an additional 120 measurements in the event the main power supply becomes depleted.

2.6.1 Battery Charger. An automatic charger with quick-disconnect plug-in type connectors shall be provided. The time it takes to fully recharge the battery shall not be greater than sixteen hours. The charger shall operate from 12 volts, dc or 110 volts, ac, 60-hertz power, and shall protect the voltage regulator from damage when the battery is connected improperly. The tester shall incorporate a battery overcharge prevention feature.

2.6.2 Rechargeable Batteries. If rechargeable batteries are used, they shall be of a type that minimizes the promotion of corrosion, when left installed for extended periods.

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2.7 Voltage indicator. An indicator or other device shall provide a clear distinction between when the voltage is adequate for testing, low, and when the TCMMD is running on back-up power.

2.8 Housing. The power supply, sources, detectors, readout device, etc., shall be in a housing of rugged construction that is moisture and dust proof. The housing shall conform to the U.S. Nuclear Regulatory Commission (NRC) requirements in Title 10 of the Code of Federal Regulations.

2.9 Tools and accessories. The TCMMD shall be furnished with a reference standard, all tools necessary for operation, site-preparation devices, technical publications, an Alternating Current (AC) battery charger, 24-V Direct Current (DC) battery charger, operating instructions, and a transport case with locking mechanism. All tools and accessories shall be stored and transported within the same case as the TCMMD. The tools shall include, at a minimum, guide-plate/scraper, drill rod/pin, drill rod/pin drive/insertion tool, drill rod/pin extraction tool, lubrication, sign kit, and computer data download cable.

2.9.1 Reference standard. The reference standard shall be a combination type, provided to check equipment operation, background count, and to establish conditions for a reproducible density and moisture count rate.

2.9.2 Technical Publications. Department of the Army Technical Manuals will be used to support this system IAW MIL-STD-40051-2 and AR 25-30. Supplemental data will include Maintenance Allocation Charts (MAC)'s, Pre-Maintenance Check (PMC), and Repair Parts Special Tools Listing (RPSTL).

2.9.3 Backscatter. Site preparation devices for the backscatter test method shall include a flat plate straightedge or other suitable tools, to be used to level the test site to the required uniformity.

2.9.4 Direct Transmission. When required, the provided site preparation devices for the direct transmission test method shall include an auger, rod, or other hole forming device with marks to indicate depth. It shall have a bore diameter equal to or up to .12 inch larger than the probe, with a guide to ensure that the hole is perpendicular to the test site surface.

2.9.5 Air Gap (Optional). The provided sire preparation devices for the (optional) air gap test method shall include a cradle or spacers, to support the gage at the optimum gap above the material being tested.

2.10 Transport case. The transport case shall contain the tester, all tools, accessories, technical publications and a locking mechanism. The case shall conform to the U.S. Department of Transportation requirements in Title 49 of the Code of Federal Regulations. The exterior of the case shall contain all labels/markings as required by the regulations.

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2.11 Performance. The TCMMD shall meet the performance requirements of ASTM Standards D6938 and D2950. The TCMMD shall also meet the following requirements.

2.11.1 Direct transmission.

Range of calibration	80 to 160 pounds per cubic foot (pcf)
Precision (6-inch depth) (at 120 pcf)	± 0.25 pcf
Chemical error (at 120 pcf)	± 1.50 pcf
Surface roughness error (0.05 in.)	– 0.90 pcf

2.11.2 Backscatter.

Range of calibration	80 to 160 pcf
Precision (6-inch depth) (at 120 pcf)	± 0.50 pcf
Chemical error (at 120 pcf)	± 2.50 pcf
Surface roughness error (0.05 in.)	– 4.00 pcf

2.11.3 Moisture.

Range of calibration	0 to 40 pcf
Precision (at 10 pcf)	± 0.25 pcf

2.12 Calibration. The TCMMD shall be enabled with a self calibration feature as well as the ability to be calibrated via the data entry of coefficients.

2.13 Operational alarm. The tester shall emit an auditory pitch or sound, not to exceed 85 dB(A) at the operator's position, when it is in its operational mode (i.e, the radioactive source is in the unshielded position) .

2.14 Environmental conditions. The TCMMD shall operate as specified herein in any ambient temperatures from 33º F to 110ºF. The TCMMD shall operate as specified herein on hot material (asphalt) at temperatures up to 350º F without the use of additional insulating material between the instrument and the surface being tested. The TCMMD shall be capable of operating in wet/damp, cold, hot, humid/tropical, dry, and dusty environmental conditions. Operation shall not be hindered by personnel wearing cold weather clothing (specifically gloves).

2.15 Leak Testing. The TCMMD will be provisioned to permit leak testing. The leak test can be performed manually by the operator or it can be an integral method of performing a self-diagnostic leak test that displays the results on the visual display.

2.16 Weight. The tester weight shall not be greater than 42 pounds. The weight of the tester and case (with tools, publications, and accessories) shall not be greater than 116 pounds.

2.17 Serviceability. Serviceable subassemblies shall be removable without exposing the sources or disturbing their shielding assemblies.

2.18 Marking. The TCMMD housing and transport case shall be marked with the unit serial and model numbers, National Stock Number (NSN), the type of radioactive sources (isotopes) and date of manufacture, the amount of radioactive material (activity), radiation symbol, source

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manufacturer's name and address, and the words "CAUTION – RADIOACTIVE MATERIAL". The manufacturer shall ensure the TCMMD has the proper radioactive material markings and labels In Accordance With (IAW) applicable U.S. and International Law are affixed to the TCMMD housing and transportation case.

2.19 General Transport Requirements. The TCMMD shall have the capability of being transportable worldwide by rail, marine, highway, and air modes. The TCMMD must be highway transportable by High Mobility Multipurpose Wheeled Vehicle (HMMWV) and Family of Medium Tactical Vehicle (FMTV). The TCMMD shall be air transportable by C-130, C-17, C-5 and CH-47 aircraft. The TCMMD shall be transportable by standard military sea and rail assets. The TCMMD shall be capable of being shipped in its transport case IAW DOT, NRC, IATA and 49 CFR Regulations.

2.20 Survivability.

2.20.1 Electronic Attack (EA) and Wartime Reserve Modes (WARM). It's desired that the TCMMD remain operational when subjected to Electrostatic Discharge (ESD), Electromagnetic Interference (EMI), Electromagnetic Compatibility (EMC), and Near Strike Lightning (NSL).

2.20.2 Nuclear, Biological, and Chemical (NBC) Contamination. It's desired that the TCMMD be able to operate in an NBC environment and survive decontamination. It's desired that the TCMMD be capable of being decontaminated to negligible risk levels, with minimum replacements of exposed components. The TCMMD shall allow full operation by personnel in full protective ensemble (MOPP level IV).

2.20.3 High-Altitude Electromagnetic Pulse. The TCMMD system upset is acceptable during a nuclear event. It's desired that the TCMMD shall be capable of being placed back into operation after a nuclear event. It's also desired that the TCMMD must be capable of performing critical operations following High-Altitude Electromagnetic Pulse (HEMP) exposure. The design shall conform to the requirements of MIL-STD-464. The HEMP environment is described in MIL-STD-2169.

3. REGULATIORY REQUIREMENTS

3.1 Licensing and control of radiation sources. The contractor shall provide evidence that the tester has met the product safety requirements of the U.S. Nuclear Regulatory Commission for products containing radioactive sources, and provide the necessary data for licensing. The contractor shall furnish the contracting officer a copy of the license(s) required prior to shipment date, to comply with the requirements of Title 49 of the Code of Federal Regulations.

3.2 Recovered material. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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3.3 Specifications and standards. The following specifications and standards form a part of this purchase description. The latest versions of these documents shall be used on the date the solicitation is released.

(Unless otherwise indicated, copies of federal/military specifications, standards, and handbooks are available from the Acquisition Streamlining and Standardization Information System (ASSIST) database (https://assist.daps.dla.mil) Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Standards

Federal

DOD 4500.9-R	Defense Transportation Regulation

(Application for copies should be addressed to the DLMSO, ATTN: DLMSO J-6251 Suite 1650, Defense Logistics Agency, J-6, 8725 John J Kingman Road, STOP 6205, Fort Belvoir, VA 22060-6217. This document may be accessed at http://www.dla.mil/j-6/dlmso/elibrary/manuals/regulations.asp)

DOD I 6055.8	Occupational Ionizing Radiation Protection Program
DOD 4160.21-M-1	Defense Demilitarization Manual

(Application for copies of the two previous documents should be addressed to the Defense Technical Information Center, 8725 John J. Kingman Rd. Suite 0944, Ft. Belvoir, VA 22060-6218. These documents may be accessed at http://www.dtic.mil/)

10 CFR	Nuclear Regulatory Commission
29 CFR	Labor
40 CFR	Protection of the Environment
49 CFR	Transportation

(The Code of Federal Regulations (CFR) are for sale on a subscription basis by the U.S. Government Printing Office, 732 N. Capitol Street, NW, Washington, DC 20401. Online access is available at http://www.gpoaccess.gov/cfr .)

Military

AR 385-10	Army Safety Program
AR 700-1	Environmental Protection and Enhancement
AR 700-2	Environmental Effects of Army Actions
AR 200-1	Environmental Protection and Enhancement

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AR 70-1	Army Acquisition Policy
AR 40-5	Preventative Medicine
AR 40-10	Health Hazard Assessment Program in Support of the Army Acquisition Process
AR 750-43	Army Test, Measurement, and Diagnostic Equipment
AR 725-50	Requisitioning, Receipt, and Issue System
AR 735-5	Policies and Procedures for Property Accountability
DA Pam 385-16	System Safety Management Guide
DA Pam 385-24	Army Radiation Safety Program
DA Pam 40-18	Personnel Dosimetry Guidance & Dose Recording Procedures for Personnel Occupationally Exposed to Ionizing Radiation

(Army Regulations and DA Pamphlets may be accessed at http://www.army.mil/usapa/)

DLAI 4145.8/NAVSUPINST 4000.34C/AFJI 23-504/MCO P4400.105 Radioactive Commodities in the Department of Defense Supply System

(This document may be accessed at http://www.dla.mil/dlaps/dlai/i4145.8.pdf)

AMC-R 11-48 Radiation Safety Program

(This document may be accessed at http://www.amc.army.mil/pa/pubs/r11_48.pdf)

MIL-STD-464	Electromagnetic Environment Effects Requirements for Systems
MIL-STD-2169 (Classified)	High Altitude Electromagnetic Pulse (HEMP) Environment

(Application for copies of MIL-STD-2169 should be addressed with a need-to-know to: HQ DTRA, ATTN: TD/TDANE, 6801 Telegraph Road, Alexandria, VA 22310-3398.)

MIL-STD-129	Military Marking for Shipping and Storage
TB 43-0116	Identification of Radioactive Materials in the Army
TB 750-25	Maintenance of Supplies and Equipment: Army Test, Measurement, and Diagnostic Equipment (TMDE) Calibration and Repair Support (C&RS) Program
TB 9-6665-285-15	Army Calibration Program for Radiac Meters
TB 43-0180	Calibration and Repair Requirements for the Maintenance of Army Materiel
TB 43-4	Safety Requirements for Maintenance of Electrical and Electronic Equipment

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TM 3-261	Handling and Disposal of Unwanted Radioactive Material
TM 38-250	Preparing Hazardous Materials for Military Air Shipments

(Technical Bulletins and Technical Manuals may be accessed at https://www.logsa.army.mil/)

3.4 Other Government Documents, Drawings and Publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Surface Deployment and Distribution Command Transportation Engineering Agency (SDDCTEA)

TEA PAMPHLET 70-1 July 2005	Transportability and Deployability for Better Strategic Mobility

(Application for copies should be addressed to the Surface Deployment and Distribution Command Transportation Engineering Agency, ATTN: MTTE-TR, 720 Thimble Shoals Blvd, Suite 130, Newport News, VA 23606-2574)

DEPARTMENT OF LABOR (DOL)

29 CFR 1910- Occupational Safety and Health Standards (OSHA)
29 CFR 1926- Safety and Health Regulation for Construction

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government printing office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof).

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3.5 Other Publications. The following document(s) form a part of this purchase description to the extent specified herein.

American Society of Testing and Materials (ASTM)

ASTM D 6938	Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)
ASTM D 2950	Standard Test Method of Density of Bituminous Concrete in Place by Nuclear Methods
ASTM D 5220	Standard Test Method for Water Mass per Unit Volume of Soil and Rock by the Neutron Depth Probe Method

(ASTM standards may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103 [http//www.ASTM.org]).

3.6 Order of Precedence. In the event of a conflict between the test of this purchase description and the references cited (except of associated detail specifications or Military Standards (MS)), the text of this purchase description shall take precedence. Nothing in this purchase description, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance as provided for in the provisions of the contract or purchase order.

4.2 Market acceptability. All TCMMD provided in accordance with this commercial item description shall be the manufacturer's current commercial production model equipped with all components furnished as standard equipment and features customarily furnished with TCMMD of this type, whether stipulated herein or not. The TCMMD shall be able to perform nondestructive density measurements of soils, asphalt pavement, and other construction materials. The TCMMD shall be able to perform non-destructive measurements of the moisture content of soils. The TCMMD shall be able to perform non-destructive measurements through nuclear backscatter, this requires a hole punched into the test material and the gamma source inserted to a specific depth , and (optional) air gap method. The TCMMD shall have been in the field at least one year by professional organizations such as State Highway Departments.

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5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 INTENDED USE. During U.S. Military operations U.S. Army Technical Engineer Teams will employ the TCMMD during the initial analytical processing of the potential U.S. Military operational site to rapidly, and precisely analyze in-situ soils and other materials to determine if the intended U.S. Military operation can take place or as to what solutions can be developed and employed to increase an existing surface to attain desired surface strength levels for enabling the specific U.S. Military operation. The TCMMD will be used at-will/on-demand to perform Quality Assurance/Quality Control (QA/QC) assessments before, during, and after a horizontal construction operation. The tester will be used to ensure as-constructed area design specifications are precisely followed.

6.2 Data requirements. The procuring agency should specify the level of preservation, packing, and marking required. The procuring agency may also need to specify inclusion of a gamma detection meter (monitor), when it is a requirement by the user's local regulatory agencies. The procuring agency should also specify that the manufacturer provides NRC Sealed Source and DOT 7A certification, radioactive material caution signs (Energy Reorganization Act of 1974 – Section 206), and the latest version of 10 CPR Parts 19, 20 and 21, the latest version of NRC Form 3 with each tester. The procuring agency should specify that the transport case be moisture and dust resistant and have corrosion resistant hardware.

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Preparing activity:
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Review Activity:
Army – AT

Project 6635-2010-001

Civil Coordinating Activity:
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